

Result 3.0: Explicit Finite Spin-Chain Hamiltonian for the SOC Workbench

July 13, 2026

1 Purpose of the test

This result tests whether the phenomenological nearest-neighbour spin-correlation input used in the projected bright-dark SOC model can be replaced by an explicit finite $S = 1/2$ spin-chain Hamiltonian. The working hypothesis is that the spin sector can be solved independently to obtain

$$C_1(T, B) = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+1} \rangle_{T, B}, \quad (1)$$

and that this value can then enter the same projected optical coupling used in the Step 3 dynamic phonon model.

2 Physical model

The spin Hamiltonian implemented in the workbench is

$$H_{\text{spin}} = \sum_i J_i(T, B) \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+2} + g\mu_B B \sum_i S_i^z, \quad (2)$$

with

$$J_i(T, B) = J [1 + (-1)^i \delta(T, B)]. \quad (3)$$

The finite chain uses $N = 6$ spins and open boundary conditions. The optical sector is kept aligned with the Step 3 notebook:

$$H_{\text{opt}} = H_{\text{orb}} \otimes I_{\text{ph}} + \omega_Q I_{\text{orb}} \otimes a^\dagger a + V_{BD} L_{BD} \otimes I_{\text{ph}} + g_Q L_{BD} \otimes (a + a^\dagger). \quad (4)$$

Here $L_{BD} = |B\rangle\langle D| + |D\rangle\langle B|$ and

$$V_{BD} = V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B). \quad (5)$$

The full spin Hilbert space is not tensored into the Liouville spectroscopy calculation in this result; it is used to compute C_1 exactly for the projected optical model.

3 Calculation definitions

The workbench is controlled by a single Python dictionary, `model_params`, which contains all physical and numerical model parameters for the active calculation. Rephasing spectra were evaluated on $\omega_1 < 0$, $\omega_3 > 0$, while unrephasing spectra were evaluated separately on $\omega_1 > 0$, $\omega_3 > 0$.

4 Active parameter set and acceptance criteria

Output stem	T (K)	B (T)	N_{spin}	Active spin source
SOC_workbench	7.0	0.0	6	exact H_{spin}

Acceptance criteria: the spin Hamiltonian must be Hermitian and have dimension 2^N ; every run must produce linked raw arrays and the six mandatory real, imaginary, and absolute rephasing/unrephasing figures; the rephasing and unrephasing frequency quadrants must be distinct; and C_1 must be computed from the explicit spin Hamiltonian and injected into V_{BD} .

5 Numerical results

The generated metrics are stored in `Analysis/step03__spin_chain_exact_metrics.csv`. The main derived values are:

Output	T	B	δ	C_1	V_{BD}	Reph. $ S $ int.
SOC_workbench	7.0	0.0	0.00707	-0.49863	0.02267	4.854e+02

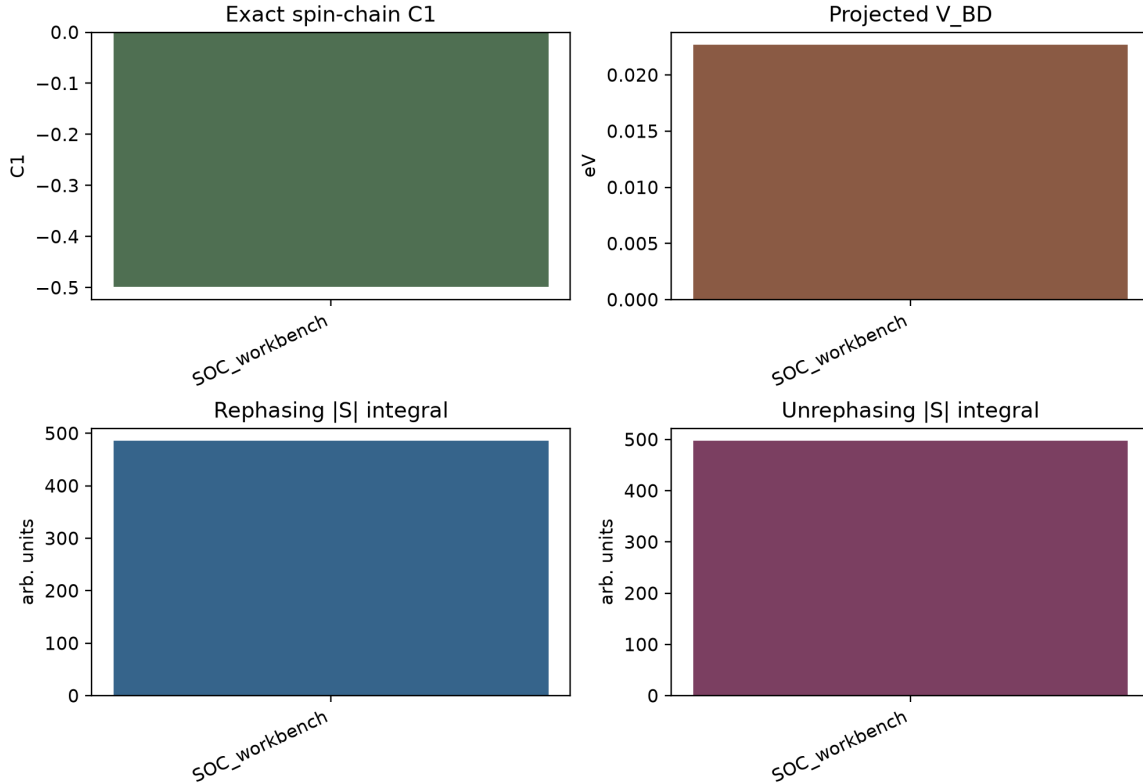


Figure 1: Exact spin-chain C_1 , projected V_{BD} , and integrated 2D response metrics for the active parameter dictionary.

6 Validation outcome

The validation details are saved in `Analysis/step03__spin_chain_exact_validation_outcome.txt`. All acceptance criteria are evaluated from raw arrays and generated file existence. The evidence supports the limited conclusion that this workbench now contains a real finite-chain H_{spin} and uses its thermal nearest-neighbour correlation in the projected optical SOC model. It does not yet validate a full tensor-product spin-orbital Liouville calculation.